

Canada Import Gap — Yiwu Small Commo Opportunity Analysis

Mondoré Consulting · Strategic Intelligence Report

Objective: Identify export opportunities in the Canadian market for products supplied by the **Yiwu wholesale market** — small consumer commodities including toys, bags, kitchenware, home décor, apparel accessories, and seasonal goods.

Methodology:

- 1 . **Macro scan** — compare Canada's total world imports vs. imports from China across all HS 2 chapters
- 2 . **Universe filter** — isolate chapters that constitute the Yiwu small commodity market
- 3 . **Gap analysis** — for each Yiwu segment: $\text{Gap} = \text{World Imports} - \text{China Imports}$
- 4 . **Opportunity Score** = $\text{Gap} \times (1 - \text{China Share}\%)$ — rewards large gap **and** low Chinese penetration
- 5 . **HS 6 deep dive** — drill into 6 -digit product codes for highest-score segments

Data source: UN Comtrade API · Canada (reporter 1 2 4) · Years 2 0 2 1 – 2 0 2 .

```
In [1]: # — Environment setup —————
import sys, os
sys.path.insert(0, os.path.dirname(os.path.abspath('__file__')))

import warnings
warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.gridspec as gridspec
from matplotlib.colors import LinearSegmentedColormap
import seaborn as sns
from IPython.display import display, HTML

# Chart style – clean white for PDF export
plt.rcParams.update({
    'figure.facecolor': 'white',
    'axes.facecolor': '#F8F9FA',
    'axes.edgecolor': '#DEE2E6',
    'axes.labelcolor': '#212529',
    'xtick.color': '#495057',
    'ytick.color': '#495057',
    'text.color': '#212529',
    'grid.color': '#DEE2E6',
    'grid.linestyle': '--',
    'grid.alpha': 0.7,
```

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    'font.family':      'DejaVu Sans',
    'font.size':        10,
    'axes.titlesize':   13,
    'axes.titleweight': 'bold',
    'axes.labelsize':   10,
    'figure.dpi':       130,
})

```

```

CANADA_RED = '#D52B1E'
CHINA_GOLD = '#F4A300'
GAP_BLUE   = '#1A73E8'
ACCENT_TEAL = '#00897B'
LIGHT_GRAY = '#E9ECEF'

```

```
print('✅ Libraries loaded')
```

✅ Libraries loaded

```

In [2]: # — Import project modules —————
from config import CHINA_CODE, WORLD_CODE, DEFAULT_YEARS
from api import fetch_hs2_all_years, fetch_hs6_chapter
from analysis import records_to_df

YEARS = [2021, 2022, 2023, 2024]

print(f'Loading data for years: {YEARS}')
print('This may take a moment on first run (API calls are cached locally)')

# — Load HS2 data – try live API, fall back to demo —————
USE_DEMO = False # set True to use synthetic demo data

try:
    world_r = fetch_hs2_all_years(WORLD_CODE, YEARS, use_demo=USE_DEMO)
    china_r = fetch_hs2_all_years(CHINA_CODE, YEARS, use_demo=USE_DEMO)
    if not world_r:
        raise ValueError('Empty response from API')
    DATA_MODE = '🌐 Live – UN Comtrade API'
except Exception as e:
    print(f'⚠️ Live API unavailable ({e}). Falling back to demo data.')
    world_r = fetch_hs2_all_years(WORLD_CODE, YEARS, use_demo=True)
    china_r = fetch_hs2_all_years(CHINA_CODE, YEARS, use_demo=True)
    DATA_MODE = '📦 Demo – Synthetic (calibrated to Statistics Canada)'

print(f'Data mode: {DATA_MODE}')
print(f'World records: {len(world_r):,}   China records: {len(china_r):,}')

```

Loading data for years: [2021, 2022, 2023, 2024]

This may take a moment on first run (API calls are cached locally)...

```

[api] 2021 partner=0 cmd=AG2 – attempt 1
[api] → 33758 records received
[api] 2022 partner=0 cmd=AG2 – attempt 1
[api] → 33967 records received
[api] 2023 partner=0 cmd=AG2 – attempt 1
[api] → 33498 records received
[api] 2024 partner=0 cmd=AG2 – attempt 1
[api] → 33999 records received
[api] 2021 partner=156 cmd=AG2 – attempt 1
[api] → 10452 records received
[api] 2022 partner=156 cmd=AG2 – attempt 1
[api] → 10971 records received
[api] 2023 partner=156 cmd=AG2 – attempt 1

```

```
[api] → 10560 records received
[api] 2024 partner=156 cmd=AG2 – attempt 1
[api] → 11190 records received
Data mode: 🌐 Live – UN Comtrade API
World records: 135,222 China records: 43,173
```

```
In [3]: # — Build HS2 summary DataFrame —————
world_df = records_to_df(world_r)
china_df = records_to_df(china_r)

world_agg = (
    world_df
    .groupby(['hs_code', 'description'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'world_usd'})
)
china_agg = (
    china_df
    .groupby(['hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'china_usd'})
)

hs2 = world_agg.merge(china_agg, on='hs_code', how='left')
hs2['china_usd'] = hs2['china_usd'].fillna(0)
hs2['gap_usd'] = hs2['world_usd'] - hs2['china_usd']
hs2['china_share_pct'] = (hs2['china_usd'] / hs2['world_usd']).replace(0,
hs2['world_b'] = (hs2['world_usd'] / 1e9).round(3)
hs2['china_b'] = (hs2['china_usd'] / 1e9).round(3)
hs2['gap_b'] = (hs2['gap_usd'] / 1e9).round(3)
hs2['opp_score'] = (hs2['gap_usd'] * (1 - hs2['china_share_pct']) /
hs2['short_desc'] = hs2['description'].str[:40]
hs2['label'] = hs2['hs_code'] + ' · ' + hs2['short_desc']
hs2 = hs2.sort_values('opp_score', ascending=False).reset_index(drop=True)
hs2.index += 1
hs2.index.name = 'rank'

print(f'HS2 chapters loaded: {len(hs2)}')
print(f'Total Canada imports (all years): USD {hs2["world_b"].sum():,.0f}')
print(f'China share (average): {(hs2["china_b"].sum() / hs2["world_b"].su
```

```
HS2 chapters loaded: 102
Total Canada imports (all years): USD 8,650 B
China share (average): 15.3%
```

Section 1 — Macro Overview: All Canadian Import

Before zooming into Yiwu categories, we scan the entire import landscape to understand the size and structure of Canada's trade deficit with China across all sectors.

```
In [4]: # — KPI Summary —————
total_world = hs2['world_b'].sum()
total_china = hs2['china_b'].sum()
total_gap = hs2['gap_b'].sum()
avg_share = total_china / total_world * 100
n_low = (hs2['china_share_pct'] < 15).sum()

fig, axes = plt.subplots(1, 5, figsize=(17, 2.5))
```

```

fig.suptitle(f'Canada Imports – All HS2 Chapters · {YEARS[0]}–{YEARS[-1]}',
            fontsize=11, y=1.02, color='#495057')

kpis = [
    ('Total World Imports', f'USD {total_world:,.0f} B', CANADA_RED),
    ('Total China Imports', f'USD {total_china:,.0f} B', CHINA_GOLD),
    ('Total Import Gap', f'USD {total_gap:,.0f} B', GAP_BLUE),
    ('China Avg Share', f'{avg_share:.1f}%', ACCENT_TEAL),
    ('Categories CN<15%', f'{n_low} chapters', '#7B1FA2'),
]

for ax, (title, val, color) in zip(axes, kpis):
    ax.set_facecolor(color)
    ax.text(0.5, 0.65, val, ha='center', va='center', fontsize=18, fontcolor='white')
    ax.text(0.5, 0.25, title, ha='center', va='center', fontsize=9, color='white')
    ax.set_xticks([]); ax.set_yticks([])
    for spine in ax.spines.values(): spine.set_visible(False)

plt.tight_layout()
plt.savefig('fig_01_kpis.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 1 saved.')

```

Canada Imports – All HS2 Chapters · 2021–2024 · [Live](#) — UN Comtrade API



Figure 1 saved.

```

In [5]: # — Figure 2: Top 25 Gaps – All HS2 —————
top25 = hs2.head(25).sort_values('world_b', ascending=True)

fig, ax = plt.subplots(figsize=(13, 10))

bars_gap = ax.barh(top25['label'], top25['gap_b'], color=GAP_BLUE,
                  left=top25['gap_b'])
bars_china = ax.barh(top25['label'], top25['china_b'], color=CHINA_GOLD,
                   left=top25['gap_b'])

for bar, row in zip(bars_gap, top25.itertuples()):
    ax.text(bar.get_width() + row.china_b + 0.3, bar.get_y() + bar.get_height() * 0.5,
            f'CN {row.china_share_pct:.0f}%', va='center', fontsize=8, color='white')

ax.set_xlabel('USD Billions (cumulative 2021–2024)', labelpad=8)
ax.set_title('Top 25 Import Categories – Canada World vs China Gap', pad=10)
ax.legend(loc='lower right', framealpha=0.9)
ax.grid(axis='x', alpha=0.5)
ax.set_axisbelow(True)
plt.tight_layout()
plt.savefig('fig_02_top25_gaps.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 2 saved.')

```

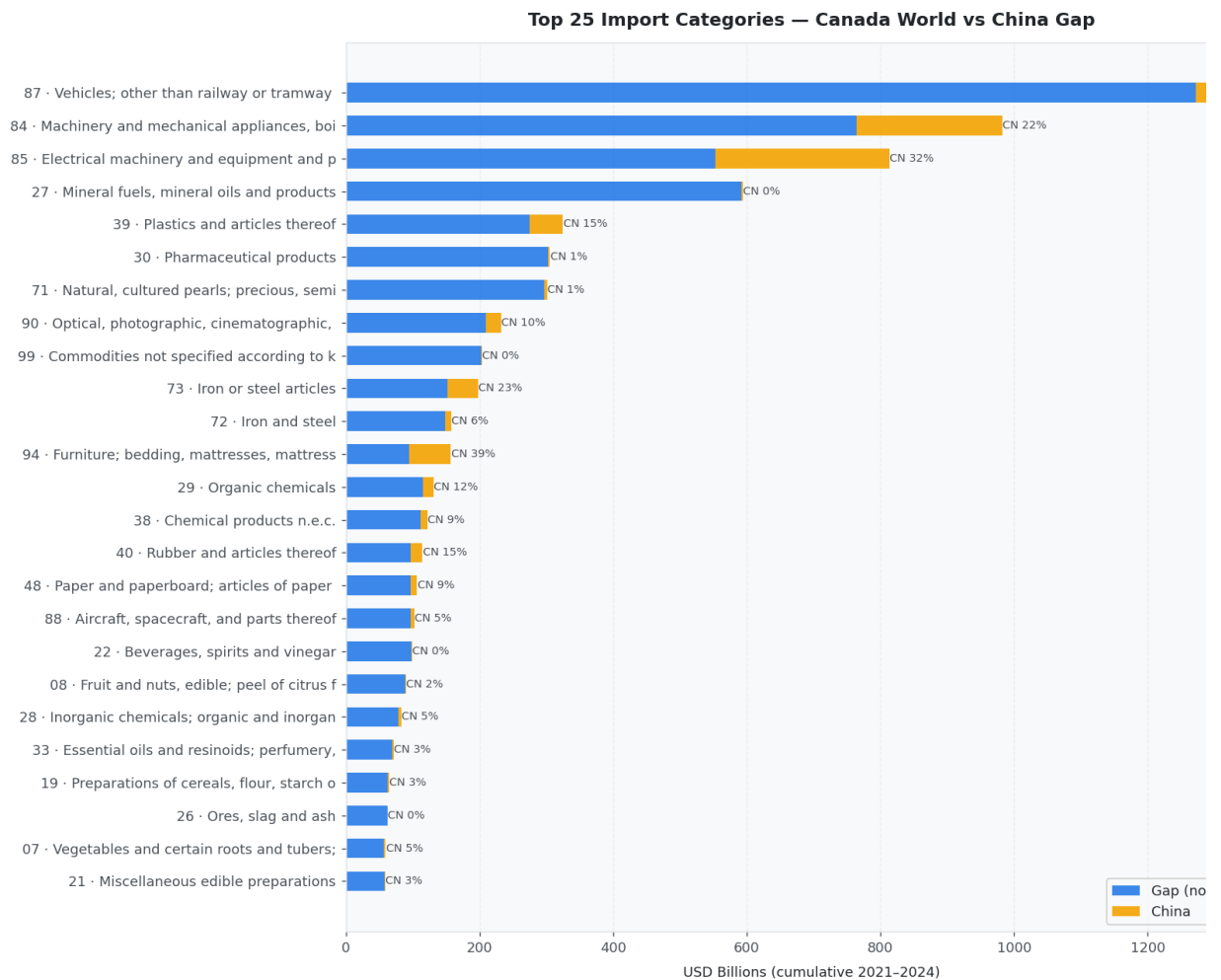


Figure 2 saved.

```
In [6]: # — Figure 3: Strategic Quadrant – All HS2 —
fig, ax = plt.subplots(figsize=(14, 9))

med_share = 25
med_gap = hs2['gap_b'].median()

scatter = ax.scatter(
    hs2['china_share_pct'], hs2['gap_b'],
    s=hs2['world_b'] * 6,
    c=hs2['opp_score'], cmap='RdYlGn_r',
    alpha=0.75, edgecolors='white', linewidth=0.4,
)

plt.colorbar(scatter, ax=ax, label='Opportunity Score (B USD)', shrink=0.

ax.axvline(med_share, color='#ADB5BD', linestyle='--', lw=1.2, label=f'Me
ax.axhline(med_gap, color='#ADB5BD', linestyle='--', lw=1.2, label=f'Me

# Quadrant labels
y_max = hs2['gap_b'].max()
quad_style = dict(fontsize=12, fontweight='bold', alpha=0.35)
ax.text(3, y_max * 0.88, '🎯 PRIORITY', color=ACCENT_TEAL, **quad_sty
ax.text(50, y_max * 0.88, '⚡ GROWTH', color=CANADA_RED, **quad_st
ax.text(3, med_gap * 0.3, '🔍 NICHE', color=GAP_BLUE, **quad_st
ax.text(50, med_gap * 0.3, '🔒 SATURATED', color='#6C757D', **quad_st

# Annotate top 10 by score
for _, row in hs2.head(10).iterrows():
```

```

ax.annotate(row['hs_code'], (row['china_share_pct'], row['gap_b']),
            fontsize=7.5, ha='center', va='bottom',
            xytext=(0, 5), textcoords='offset points', color='#212529')

ax.set_xlabel('China Share of Canadian Imports (%)')
ax.set_ylabel('Import Gap – World minus China (USD Billions)')
ax.set_title('Strategic Quadrant – All HS2 Chapters · Canada Import Gap A')
ax.legend(fontsize=9, loc='upper right')
ax.set_xlim(-2, 80)
ax.grid(True, alpha=0.4)
plt.tight_layout()
plt.savefig('fig_03_quadrant_all.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 3 saved.')

```

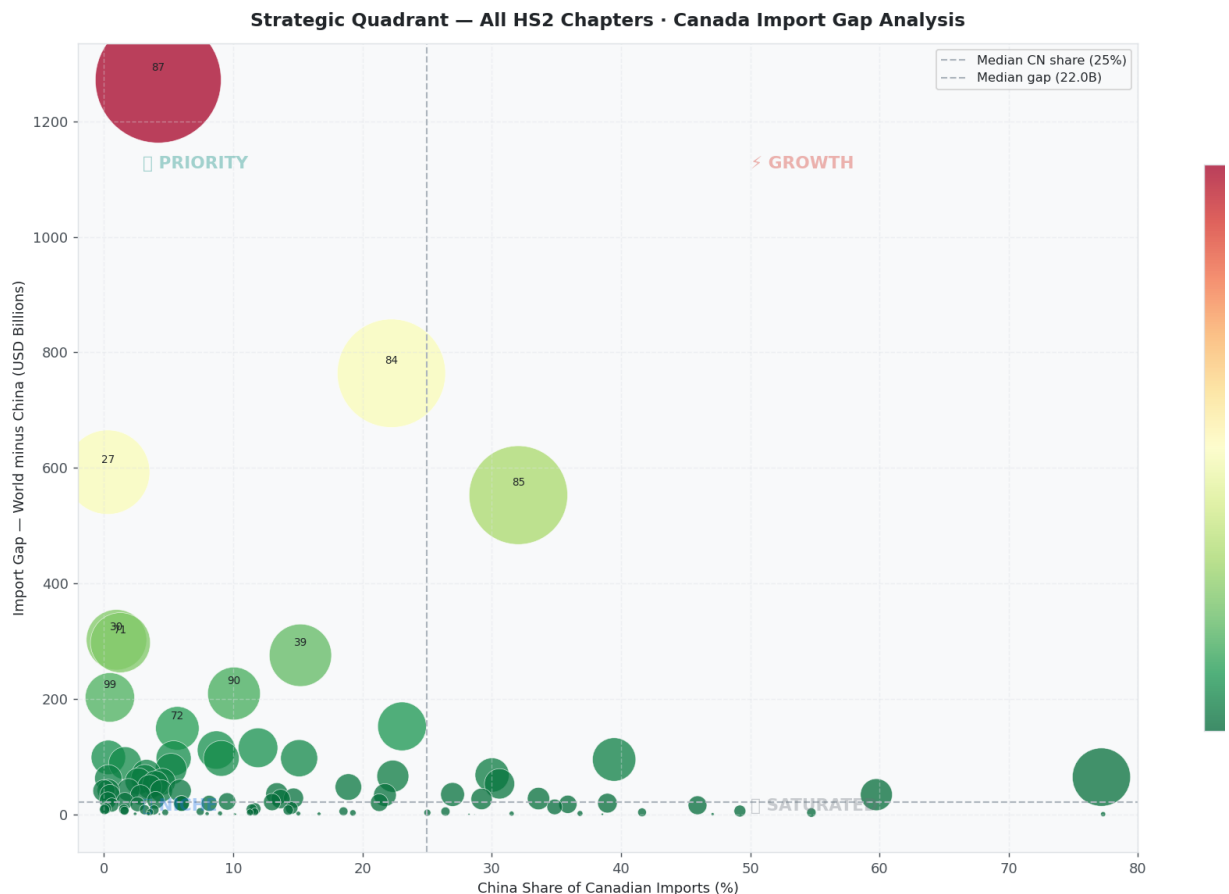


Figure 3 saved.

Section 2 — Yiwu Small Commodity Universe

The **Yiwu wholesale market** (义乌) is the world's largest small commodity trading hub, supplying 1.8 million product types across 75,000+ booths. Its core export strengths align with the following HS 2 chapters:

Segment	HS Chapters	Products
Bags & Travel Goods	42	Handbags, backpacks, wallets, luggage
Kitchenware & Tableware	39, 69, 70, 73, 82	Plastic containers, ceramics, glassware, steel cutlery

Segment	HS Chapters	Products
Home Décor & Hardware	8 3, 9 4	Frames, hooks, locks, lamps, c furniture
Textiles & Apparel	5 7, 6 1, 6 2, 6 3, 6 4, 6 5, 6 6	Carpets, clothing, towels, shoe umbrellas
Toys, Games & Sports	9 5	Toys, puzzles, outdoor sports equipment
Seasonal & Festive	4 6, 6 7, 9 6	Artificial flowers, wicker, station zippers

```
In [7]: # — Yiwu Universe Definition —————
YIWU_SEGMENTS = {
    'Bags & Travel Goods':      ['42'],
    'Kitchenware & Tableware':  ['39', '69', '70', '73', '82'],
    'Home Décor & Hardware':    ['83', '94'],
    'Textiles & Apparel':       ['57', '61', '62', '63', '64', '65', '66'],
    'Toys, Games & Sports':     ['95'],
    'Seasonal & Festive Goods': ['46', '67', '96'],
}

# Create chapter → segment map
chapter_to_segment = {}
for seg, chapters in YIWU_SEGMENTS.items():
    for ch in chapters:
        chapter_to_segment[ch.zfill(2)] = seg

YIWU_CHAPTERS = list(chapter_to_segment.keys())

# Filter HS2 to Yiwu universe
yiwu = hs2[hs2['hs_code'].isin(YIWU_CHAPTERS)].copy()
yiwu['segment'] = yiwu['hs_code'].map(chapter_to_segment)
yiwu = yiwu.sort_values('opp_score', ascending=False).reset_index(drop=True)
yiwu.index += 1
yiwu.index.name = 'rank'

print(f'Yiwu universe: {len(yiwu)} HS2 chapters across {len(YIWU_SEGMENTS)} segments')
print(f'Total Yiwu-relevant world imports: USD {yiwu["world_b"].sum():.1f} B')
print(f'China currently captures: USD {yiwu["china_b"].sum():.1f} B')
print(f'Remaining gap (opportunity): USD {yiwu["gap_b"].sum():.1f} B')
print(f'Average China share in these categories: {(yiwu["china_b"].sum() / yiwu["world_b"].sum()) * 100}%')
```

Yiwu universe: 19 HS2 chapters across 6 segments
Total Yiwu-relevant world imports: USD 1,240.6 B
China currently captures: USD 359.1 B
Remaining gap (opportunity): USD 881.6 B
Average China share in these categories: 28.9%

```
In [8]: # — Figure 4: Yiwu Universe – Segment Summary —————
seg_summary = (
    yiwu.groupby('segment', as_index=False)
        .agg(world_b=('world_b', 'sum'), china_b=('china_b', 'sum'),
             gap_b=('gap_b', 'sum'), opp_score=('opp_score', 'sum'))
)
seg_summary['share_pct'] = (seg_summary['china_b'] / seg_summary['world_b']) * 100
seg_summary = seg_summary.sort_values('opp_score', ascending=False).reset_index()

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
```

```

# Left: stacked bar by segment
ax = axes[0]
segs = seg_summary['segment']
x = np.arange(len(segs))
b1 = ax.bar(x, seg_summary['gap_b'], color=GAP_BLUE, label='Gap (non-')
b2 = ax.bar(x, seg_summary['china_b'], color=CHINA_GOLD, label='China sha
        bottom=seg_summary['gap_b'])
for i, (g, c, s) in enumerate(zip(seg_summary['gap_b'], seg_summary['chin
    ax.text(i, g + c + 0.3, f'CN {s:.0f}% ', ha='center', va='bottom', fon
ax.set_xticks(x)
ax.set_xticklabels([s.replace(' & ', '\n& ') for s in segs], fontsize=8.5
ax.set_ylabel('USD Billions (2021–2024 cumulative)')
ax.set_title('Yiwu Segments – World Import Volume', pad=10)
ax.legend()
ax.grid(axis='y', alpha=0.5)
ax.set_axisbelow(True)

# Right: opportunity score pie
ax2 = axes[1]
colors_pie = [CANADA_RED, CHINA_GOLD, GAP_BLUE, ACCENT_TEAL, '#7B1FA2', '
wedges, texts, autotexts = ax2.pie(
    seg_summary['opp_score'],
    labels=seg_summary['segment'],
    colors=colors_pie,
    autopct='%1.1f%%',
    startangle=120,
    pctdistance=0.75,
    labeldistance=1.08,
    textprops={'fontsize': 8.5},
    wedgeprops={'linewidth': 1.5, 'edgecolor': 'white'},
)
for at in autotexts: at.set_fontsize(8)
ax2.set_title('Share of Total Opportunity Score\nby Yiwu Segment', pad=10

plt.suptitle('Yiwu Small Commodity Universe – Canada Import Opportunity',
plt.tight_layout()
plt.savefig('fig_04_yiwu_segments.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 4 saved.')

```

Yiwu Small Commodity Universe – Canada Import Opportunity

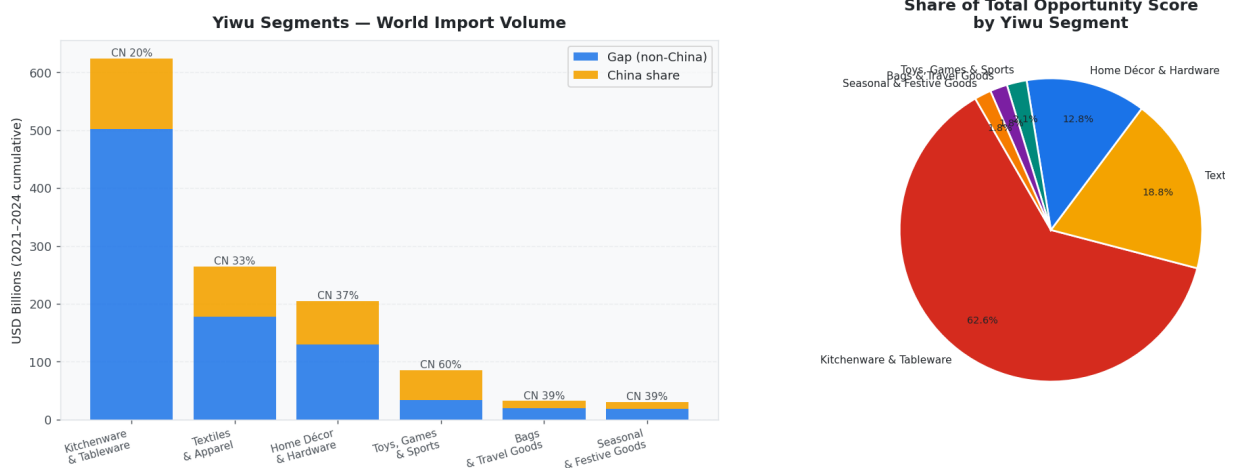


Figure 4 saved.

In [9]:


```

# — Figure 5: Yiwu Strategic Quadrant —
SEGMENT_COLORS = {
    'Bags & Travel Goods': CANADA_RED,
    'Kitchenware & Tableware': CHINA_GOLD,
    'Home Décor & Hardware': GAP_BLUE,
    'Textiles & Apparel': ACCENT_TEAL,
    'Toys, Games & Sports': '#7B1FA2',
    'Seasonal & Festive Goods': '#F57C00',
}

fig, ax = plt.subplots(figsize=(14, 8))

for seg, grp in yiwu.groupby('segment'):
    ax.scatter(
        grp['china_share_pct'], grp['gap_b'],
        s=grp['world_b'] * 18,
        c=SEGMENT_COLORS.get(seg, '#999'),
        alpha=0.78, edgecolors='white', linewidth=0.6,
        label=seg, zorder=3,
    )
    for _, row in grp.iterrows():
        ax.annotate(
            f"HS {row['hs_code']}\n{row['description'][:22]}",
            (row['china_share_pct'], row['gap_b']),
            fontsize=7.2, ha='left', va='bottom',
            xytext=(5, 5), textcoords='offset points', color='#343A40',
        )

med_share_y = yiwu['china_share_pct'].median()
med_gap_y = yiwu['gap_b'].median()
ax.axvline(med_share_y, color='#ADB5BD', linestyle='--', lw=1.2)
ax.axhline(med_gap_y, color='#ADB5BD', linestyle='--', lw=1.2)

y_max2 = yiwu['gap_b'].max()
ax.text(2, y_max2 * 0.92, '🎯 PRIORITY\n(large gap, low CN share)', for
ax.text(60, y_max2 * 0.92, '⚡ GROWTH\n(large gap, CN present)', fc
ax.text(2, med_gap_y*0.15, '🔍 NICHE\n(small gap, low CN)', fc
ax.text(60, med_gap_y*0.15, '🔒 SATURATED', f

ax.set_xlabel('China Share of Canadian Imports (%)', fontsize=11)
ax.set_ylabel('Import Gap – World minus China (USD Billions)', fontsize=1
ax.set_title('Strategic Quadrant – Yiwu Small Commodity Universe', pad=12
ax.legend(title='Segment', bbox_to_anchor=(1.01, 1), loc='upper left', fo
ax.set_xlim(-2, 90)
ax.grid(True, alpha=0.35)
plt.tight_layout()
plt.savefig('fig_05_yiwu_quadrant.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 5 saved.')

```

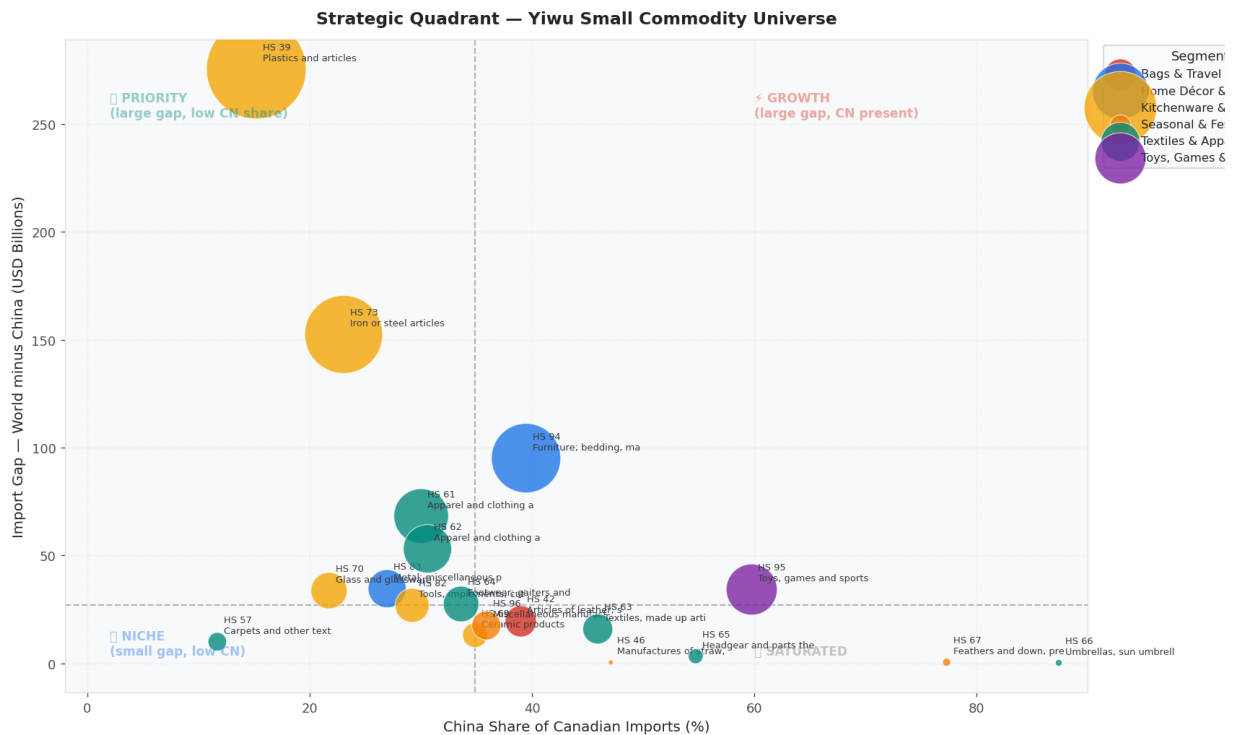


Figure 5 saved.

Section 3 — Segment Deep Dive

For each of the 6 Yiwu segments, we show: market size, China share, gap size, and oppo score per HS 2 chapter.

```
In [10]: # — Figure 6: Per-Segment Breakdown — 6-panel grid —
segments_ordered = seg_summary['segment'].tolist()
n_seg = len(segments_ordered)
ncols = 3
nrows = (n_seg + ncols - 1) // ncols

fig, axes = plt.subplots(nrows, ncols, figsize=(18, nrows * 5.5))
axes_flat = axes.flatten()

for idx, seg in enumerate(segments_ordered):
    ax = axes_flat[idx]
    grp = yiwu[yiwu['segment'] == seg].sort_values('gap_b', ascending=True)
    color = SEGMENT_COLORS.get(seg, '#888')

    b_gap = ax.barh(grp['label'], grp['gap_b'], color=color, alp
    b_china = ax.barh(grp['label'], grp['china_b'], color=CHINA_GOLD, alp
                    left=grp['gap_b'])

    for bar, row in zip(b_gap, grp.itertuples()):
        total = row.gap_b + row.china_b
        ax.text(total + 0.05, bar.get_y() + bar.get_height()/2,
                f'CN {row.china_share_pct:.0f}% · Score {row.opp_score:.2
                va='center', fontsize=7.5, color='#495057')

    ax.set_title(seg, pad=8, color=color, fontsize=11)
    ax.set_xlabel('USD Billions')
    ax.legend(fontsize=8, loc='lower right')
```

```

ax.grid(axis='x', alpha=0.4)
ax.set_axisbelow(True)

# hide empty panels
for j in range(n_seg, len(axes_flat)):
    axes_flat[j].set_visible(False)

plt.suptitle('Yiwu Segments — HS2 Chapter Detail (Gap vs China Share)', f
plt.tight_layout()
plt.savefig('fig_06_segment_detail.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 6 saved.')

```

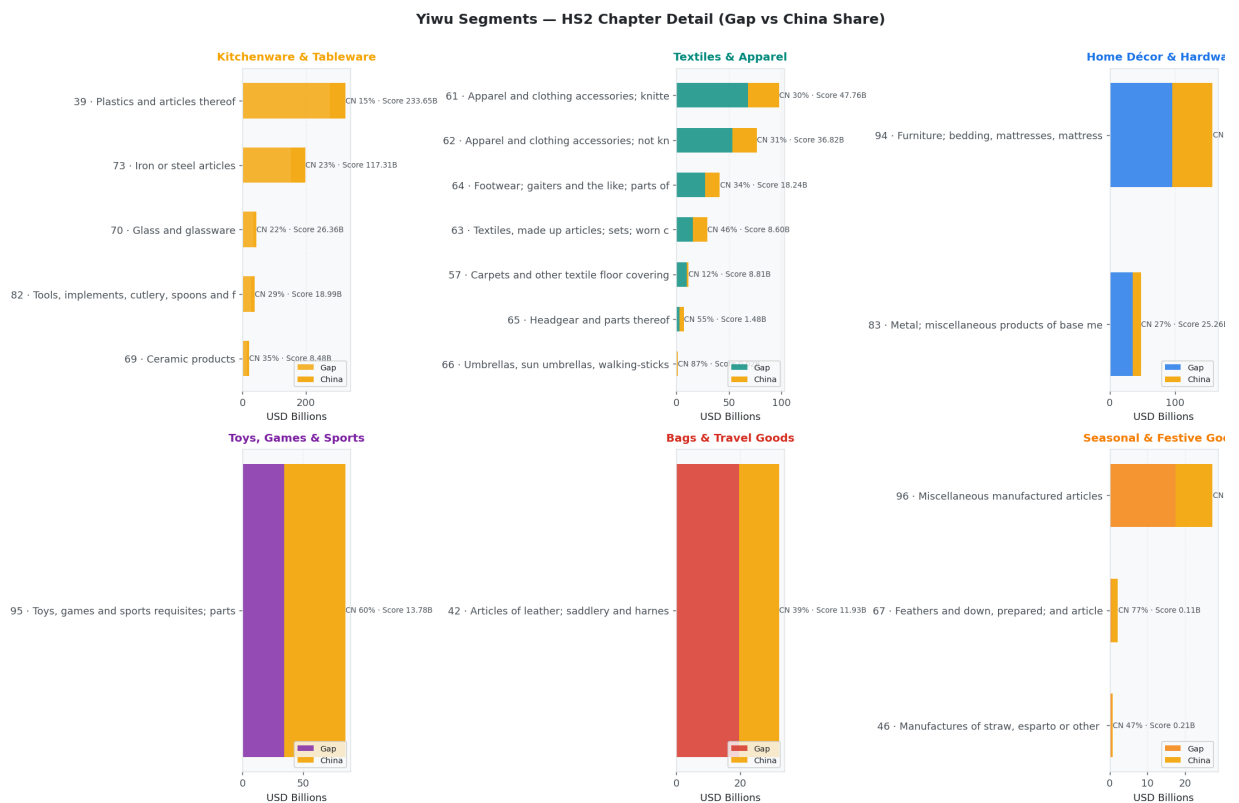


Figure 6 saved.

Section 4 — HS 6 Deep Dive: Top 3 Opportunity Chapters

We drill into 6-digit product codes for the top 3 HS 2 chapters by opportunity score to find specific products with the highest potential.

```

In [11]: # — Load HS6 data for top 3 Yiwu chapters
top3_chapters = yiwu.head(3)['hs_code'].tolist()
print(f'Loading HS6 data for chapters: {top3_chapters}')

hs6_data = {}
for ch in top3_chapters:
    try:
        w6 = fetch_hs6_chapter(WORLD_CODE, ch, YEARS, use_demo=USE_DEMO)
        c6 = fetch_hs6_chapter(CHINA_CODE, ch, YEARS, use_demo=USE_DEMO)
        if not w6: raise ValueError('Empty')
    except Exception as e:

```

```

print(f' ⚠️ HS6 live API failed for ch {ch}: {e} - using demo')
w6 = fetch_hs6_chapter(WORLD_CODE, ch, YEARS, use_demo=True)
c6 = fetch_hs6_chapter(CHINA_CODE, ch, YEARS, use_demo=True)

w6df = records_to_df(w6)
c6df = records_to_df(c6)

w6a = w6df.groupby(['hs_code', 'description'], as_index=False)['value_
c6a = c6df.groupby(['hs_code'], as_index=False)['value_usd'].sum().re

m6 = w6a.merge(c6a, on='hs_code', how='left')
m6['china_usd'] = m6['china_usd'].fillna(0)
m6['gap_usd'] = m6['world_usd'] - m6['china_usd']
m6['share_pct'] = (m6['china_usd'] / m6['world_usd']).replace(0, np.na
m6['world_m'] = (m6['world_usd'] / 1e6).round(2)
m6['china_m'] = (m6['china_usd'] / 1e6).round(2)
m6['gap_m'] = (m6['gap_usd'] / 1e6).round(2)
m6['opp_score'] = (m6['gap_usd'] * (1 - m6['share_pct'] / 100) / 1e9).
m6 = m6[m6['world_m'] > 0].sort_values('opp_score', ascending=False).
m6.index += 1; m6.index.name = 'rank'
hs6_data[ch] = m6
print(f' Chapter {ch}: {len(m6)} products loaded')

```

Loading HS6 data for chapters: ['39', '73', '94']

```

[api] 2021 partner=0 cmd=39 - attempt 1
[api] → 635 records received
[api] 2022 partner=0 cmd=39 - attempt 1
[api] → 624 records received
[api] 2023 partner=0 cmd=39 - attempt 1
[api] → 612 records received
[api] 2024 partner=0 cmd=39 - attempt 1
[api] → 604 records received
[api] 2021 partner=156 cmd=39 - attempt 1
[api] → 314 records received
[api] 2022 partner=156 cmd=39 - attempt 1
[api] → 326 records received
[api] 2023 partner=156 cmd=39 - attempt 1
[api] → 289 records received
[api] 2024 partner=156 cmd=39 - attempt 1
[api] → 313 records received
Chapter 39: 1 products loaded
[api] 2021 partner=0 cmd=73 - attempt 1
[api] → 586 records received
[api] 2022 partner=0 cmd=73 - attempt 1
[api] → 563 records received
[api] 2023 partner=0 cmd=73 - attempt 1
[api] → 576 records received
[api] 2024 partner=0 cmd=73 - attempt 1
[api] → 556 records received
[api] 2021 partner=156 cmd=73 - attempt 1
[api] → 289 records received
[api] 2022 partner=156 cmd=73 - attempt 1
[api] → 301 records received
[api] 2023 partner=156 cmd=73 - attempt 1
[api] → 286 records received
[api] 2024 partner=156 cmd=73 - attempt 1
[api] → 288 records received
Chapter 73: 1 products loaded
[api] 2021 partner=0 cmd=94 - attempt 1

```

```

[api] → 513 records received
[api] 2022 partner=0 cmd=94 – attempt 1
[rate-limit] 429 received, waiting 60s before retry...
[api] 2022 partner=0 cmd=94 – attempt 2
[api] → 525 records received
[api] 2023 partner=0 cmd=94 – attempt 1
[api] → 491 records received
[api] 2024 partner=0 cmd=94 – attempt 1
[api] → 508 records received
[api] 2021 partner=156 cmd=94 – attempt 1
[api] → 249 records received
[api] 2022 partner=156 cmd=94 – attempt 1
[api] → 251 records received
[api] 2023 partner=156 cmd=94 – attempt 1
[api] → 234 records received
[api] 2024 partner=156 cmd=94 – attempt 1
[api] → 243 records received
Chapter 94: 1 products loaded

```

```

In [12]: # — Figure 7: HS6 Deep Dive – Top 3 Chapters —————
fig, axes = plt.subplots(3, 2, figsize=(18, 20))

for row_idx, ch in enumerate(top3_chapters):
    df6 = hs6_data[ch]
    ch_name = yiwu[yiwu['hs_code'] == ch]['description'].iloc[0]
    color = SEGMENT_COLORS.get(yiwu[yiwu['hs_code'] == ch]['segment'].ilo

    top15 = df6.head(15).sort_values('world_m', ascending=True)

    # Left panel: bar chart
    ax_bar = axes[row_idx][0]
    ax_bar.barh(top15['description'].str[:45], top15['gap_m'], color=co
    ax_bar.barh(top15['description'].str[:45], top15['china_m'], color=CH
                left=top15['gap_m'])
    for bar, (_, r) in zip(ax_bar.patches[:len(top15)], top15.iterrows()):
        total_m = r['gap_m'] + r['china_m']
        ax_bar.text(total_m + 1, bar.get_y() + bar.get_height()/2,
                    f"CN {r['share_pct']:.0f}%", va='center', fontsize=7.
    ax_bar.set_title(f'HS {ch} – {ch_name[:50]}\nTop 15 Products by Marke
    ax_bar.set_xlabel('USD Millions')
    ax_bar.legend(fontsize=8)
    ax_bar.grid(axis='x', alpha=0.4); ax_bar.set_axisbelow(True)

    # Right panel: scatter opportunity
    ax_sc = axes[row_idx][1]
    sc = ax_sc.scatter(
        df6['share_pct'], df6['gap_m'],
        s=df6['world_m'] * 0.8,
        c=df6['opp_score'], cmap='YlOrRd',
        alpha=0.75, edgecolors='white', linewidth=0.5,
    )
    plt.colorbar(sc, ax=ax_sc, label='Opp Score (B)', shrink=0.8)
    # label top 5
    for _, r in df6.head(5).iterrows():
        ax_sc.annotate(r['hs_code'], (r['share_pct'], r['gap_m']),
                      fontsize=7.5, xytext=(4,4), textcoords='offset poi
    ax_sc.set_xlabel('China Share (%)')
    ax_sc.set_ylabel('Gap (USD Millions)')
    ax_sc.set_title(f'HS {ch} – Product Opportunity Map', color=color, pa
    ax_sc.grid(True, alpha=0.35)

```

```
plt.suptitle('HS6 Deep Dive – Top 3 Yiwu Opportunity Chapters', fontsize=
plt.tight_layout()
plt.savefig('fig_07_hs6_deepdive.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 7 saved.')
```



Figure 7 saved.

Section 5 — Final Opportunity Ranking: Yiwu Smart Commodities

```
# — Figure 8: Opportunity Score Ranking – Yiwu Only —
yiwu_sorted = yiwu.sort_values('opp_score', ascending=True)

fig, ax = plt.subplots(figsize=(14, max(6, len(yiwu) * 0.55)))

bar_colors = [SEGMENT_COLORS.get(s, '#888') for s in yiwu_sorted['segment']]
bars = ax.barh(yiwu_sorted['label'], yiwu_sorted['opp_score'], color=bar_colors)

for bar, (_, row) in zip(bars, yiwu_sorted.iterrows()):
    ax.text(bar.get_width() + 0.05,
            bar.get_y() + bar.get_height()/2,
            f"Score {row['opp_score']:.2f}B | CN {row['china_share_pct']}"
            va='center', fontsize=8, color='#343A40')

# Legend
legend_patches = [mpatches.Patch(color=c, label=s) for s, c in SEGMENT_COLORS.items()]
ax.legend(handles=legend_patches, title='Segment', loc='lower right', fontstyle='italic')

ax.set_xlabel('Opportunity Score (USD Billions)', fontsize=11)
ax.set_title('Yiwu Small Commodity Opportunity Ranking – Canada Import Gap'
            'Score = Gap × (1 – China Share%) · Higher = Larger Gap AND Lower Chinese Penetration'
            pad=12, fontsize=12)
ax.grid(axis='x', alpha=0.4)
ax.set_axisbelow(True)
plt.tight_layout()
plt.savefig('fig_08_yiwu_ranking.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 8 saved.')
```

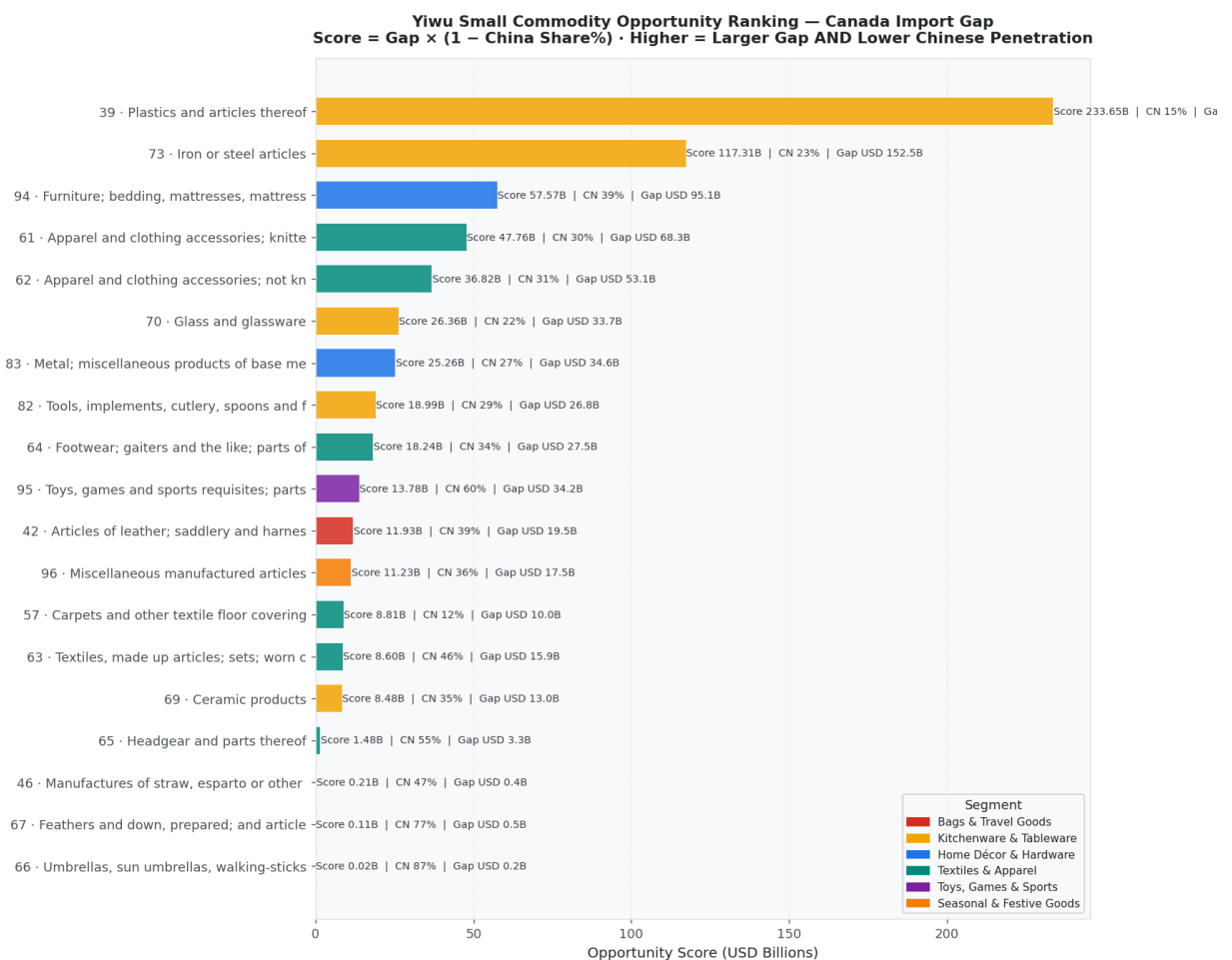


Figure 8 saved.

```

In [14]: # — Figure 9: Year-over-Year Trend – Top 5 Yiwu Chapters —————
world_yr = (
    world_df
    .groupby(['year', 'hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'world_usd'})
)
china_yr = (
    china_df
    .groupby(['year', 'hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'china_usd'})
)
trend = world_yr.merge(china_yr, on=['year', 'hs_code'], how='left')
trend['china_usd'] = trend['china_usd'].fillna(0)
trend['share_pct'] = (trend['china_usd'] / trend['world_usd']).replace(0,
trend['gap_m']      = (trend['world_usd'] - trend['china_usd']) / 1e6
trend['world_b']    = trend['world_usd'] / 1e9
trend['year']       = trend['year'].astype(int)

top5_ch = yiwu.head(5)['hs_code'].tolist()
trend5 = trend[trend['hs_code'].isin(top5_ch)].copy()
trend5 = trend5.merge(yiwu[['hs_code', 'description', 'segment']].reset_in
trend5['label'] = 'HS ' + trend5['hs_code'] + ' · ' + trend5['description']

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))

for lab, grp in trend5.groupby('label'):
    ch_code = grp['hs_code'].iloc[0]
    seg      = grp['segment'].iloc[0]
    color     = SEGMENT_COLORS.get(seg, '#888')
    grp_s = grp.sort_values('year')
    ax1.plot(grp_s['year'], grp_s['world_b'], marker='o', color=color, li
    ax2.plot(grp_s['year'], grp_s['share_pct'], marker='s', color=color,

ax1.set_title('World Import Volume (USD Billions) – Top 5 Yiwu Chapters',
ax1.set_xlabel('Year'); ax1.set_ylabel('USD Billions')
ax1.legend(fontsize=7.5); ax1.grid(True, alpha=0.4)

ax2.set_title("China's Share of Canada Imports (%) – Top 5 Yiwu Chapters"
ax2.set_xlabel('Year'); ax2.set_ylabel('China Share (%)')
ax2.legend(fontsize=7.5); ax2.grid(True, alpha=0.4)

plt.suptitle('Year-over-Year Trends (2021–2024)', fontsize=13, fontweight
plt.tight_layout()
plt.savefig('fig_09_trends.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 9 saved.')

```

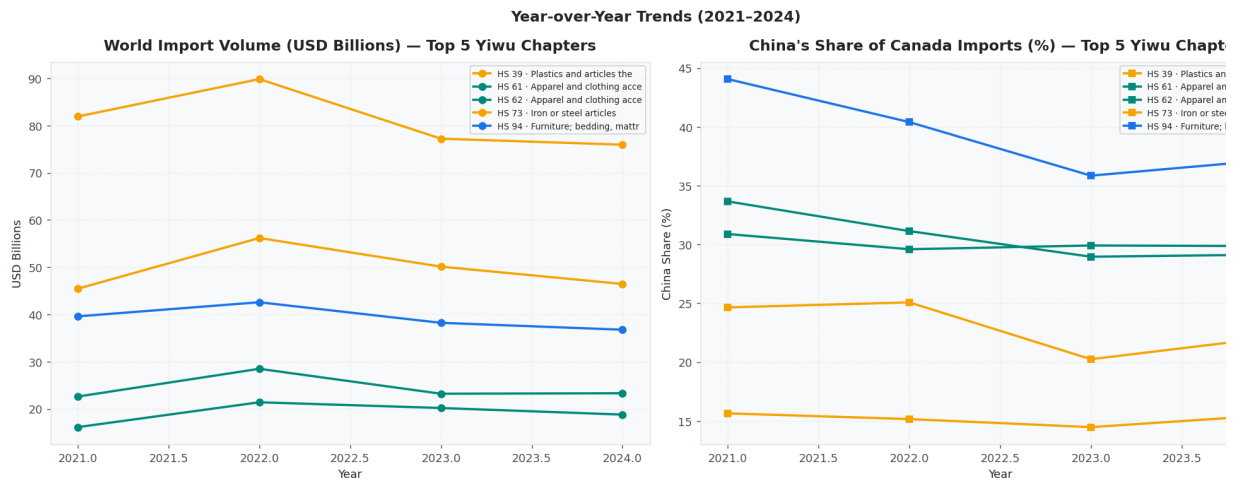



Figure 9 saved.

```
In [15]: # — Final Ranked Table
final_table = (
    yiwu.reset_index()[['rank', 'hs_code', 'description', 'segment',
                        'world_b', 'china_b', 'gap_b', 'china_share_pct', 'o
                        .copy()
    )
    final_table.columns = ['Rank', 'HS2', 'Description', 'Segment',
                          'World(B)', 'China(B)', 'Gap(B)', 'CN Share%', 'Score

print('=== FINAL YIWU OPPORTUNITY RANKING — CANADA IMPORT GAP ===')
print(final_table.to_string(index=False))

# Save to Excel for distribution
final_table.to_excel('yiwu_canada_opportunity.xlsx', index=False, sheet_n
print('\n✅ Saved: yiwu_canada_opportunity.xlsx')
```

=== FINAL YIWU OPPORTUNITY RANKING – CANADA IMPORT GAP ===

Rank
HS2

China(B)	Gap(B)	CN Share%	Score(B)	Description	Segment	World
1						
39						
				Plastics and articles thereof	Kitchenware & Tableware	324.
49.380	275.531	15.20	233.6504			
2						
73						
				Iron or steel articles	Kitchenware & Tableware	198.
45.706	152.474	23.06	117.3132			
3	94			Furniture; bedding, mattresses, mattress supports, cushions and similar stuffed furnishings; lamps and lighting fittings, n.e.c.; illuminated signs, illuminated name-plates and the like; prefabricated buildings	Home Décor & Hardware	
	157.122	62.013	95.109			
4						
61						
				and clothing accessories; knitted or crocheted	Textiles & Apparel	Ap
97.564	29.303	68.261	30.03	47.7623		
5						
62						
				clothing accessories; not knitted or crocheted	Textiles & Apparel	Appare
76.456	23.399	53.057	30.60	36.8216		
6						
70						
				Glass and glassware	Kitchenware & Tableware	
43.033	9.354	33.679	21.74	26.3568		
7						
83						
				Metal; miscellaneous products of base metal	Home Décor & Hardware	47.
12.771	34.589	26.97	25.2603			
8						
82						
				Tools, implements, cutlery, spoons and forks, of base metal; parts thereof, of base metal	Kitchenware & Tableware	
37.909	11.076	26.833	29.22	18.9923		
9						
64						
				Footwear; gaiters and the like; parts of such articles	Textiles & Apparel	
41.393	13.915	27.478	33.62	18.2400		
10						
95						
				sports requisites; parts and accessories thereof	Toys, Games & Sports	Toys, games
85.108	50.864	34.244	59.76	13.7799		
11	42			Articles of leather; saddlery and harness; travel goods, handbags and similar containers; articles of animal gut (other than silk-worm gut)	Bags & Trunks	
	32.022	12.474	19.548	38.95	11.9339	
12						

96

Miscellaneous manufactured articles Seasonal & Festive Goods
 27.322 9.807 17.516 35.89 11.2293
 13
 57

Carpets and other textile floor coverings Textiles & Apparel
 11.302 1.321 9.980 11.69 8.8136
 14
 63

worn clothing and worn textile articles; rags Textiles, made up articles;
 Textiles & Apparel 29.
 13.500 15.900 45.92 8.5988
 15
 69

Ceramic products Kitchenware & Tableware
 19.992 6.971 13.021 34.87 8.4807
 16
 65

Headgear and parts thereof Textiles & Apparel
 7.208 3.945 3.263 54.73 1.4770
 17
 46

Manufactures of straw, esparto or
 plaiting materials; basketware and wickerwork Seasonal & Festive Goods
 0.768 0.362 0.406 47.09 0.2149
 18
 67

Feathers and down, prepared; and articles made of feather or
 down; artificial flowers; articles of human hair Seasonal & Festive Goods
 2.110 1.631 0.479 77.30 0.1087
 19
 66

Umbrellas, sun umbrellas, walking-sticks,
 sticks, whips, riding crops; and parts thereof Textiles & Apparel
 1.471 1.285 0.186 87.39 0.0234

✓ Saved: yiwu_canada_opportunity.xlsx

```
In [16]: # — Styled display for notebook —————
def highlight_score(val):
    if isinstance(val, float):
        if val >= final_table['Score(B)'].quantile(0.75):
            return 'background-color: #d4edda; font-weight: bold'
        elif val >= final_table['Score(B)'].quantile(0.5):
            return 'background-color: #fff3cd'
        return ''
    return ''

def highlight_segment(val):
    seg_colors_css = {
        'Toys, Games & Sports': '#e8d5f5',
        'Home Décor & Hardware': '#d5e8f5',
        'Kitchenware & Tableware': '#ffeeba',
        'Bags & Travel Goods': '#f8d7da',
        'Textiles & Apparel': '#d4edda',
        'Seasonal & Festive Goods': '#d1ecf1',
    }
    return seg_colors_css.get(val, '')
```

```

return f"background-color: {seg_colors_css.get(val, 'white')}}"

styled = (
    final_table.style
    .applymap(highlight_score, subset=['Score(B)'])
    .applymap(highlight_segment, subset=['Segment'])
    .format({'World(B)': '{:.2f}', 'China(B)': '{:.2f}', 'Gap(B)': '{:.2f}',
            'CN Share%': '{:.1f}%', 'Score(B)': '{:.3f}'})
    .set_caption('Yiwu Small Commodity Opportunity Ranking – Canada Import')
    .set_table_styles([
        {
            'selector': 'caption',
            'props': [('font-size', '13px'), ('font-weight', 'bold'), ('color', 'red')]
        }
    ])
)
display(styled)

```

Yiwu Small Commodity Opportunity Ranking — Canada Import Gap Analysis

Rank		HS 2	Description	Segment	World(B)	China(B)	Gap(B)
0	1	3 9	Plastics and articles thereof	Kitchenware & Tableware	3 2 4 . 9 1	4 9 . 3 8	2 7 5 . 5 3 1
1	2	7 3	Iron or steel articles	Kitchenware & Tableware	1 9 8 . 1 8	4 5 . 7 1	1 5 2 . 4 7 2
2	3	9 4	Furniture; bedding, mattresses, mattress supports, cushions and similar stuffed furnishings; lamps and lighting fittings, n.e.c.; illuminated signs, illuminated name-plates and the like; prefabricated buildings	Home Décor & Hardware	1 5 7 . 1 2	6 2 . 0 1	9 5 . 1 1 3
3	4	6 1	Apparel and clothing accessories; knitted or crocheted	Textiles & Apparel	9 7 . 5 6	2 9 . 3 0	6 8 . 2 6 3
4	5	6 2	Apparel and clothing accessories; not knitted or crocheted	Textiles & Apparel	7 6 . 4 6	2 3 . 4 0	5 3 . 0 6 3
5	6	7 0	Glass and glassware	Kitchenware & Tableware	4 3 . 0 3	9 . 3 5	3 3 . 6 8 2
6	7	8 3	Metal; miscellaneous products of base metal	Home Décor & Hardware	4 7 . 3 6	1 2 . 7 7	3 4 . 5 9 2
7	8	8 2	Tools, implements, cutlery, spoons and forks, of base metal; parts thereof, of base metal	Kitchenware & Tableware	3 7 . 9 1	1 1 . 0 8	2 6 . 8 3 2
8	9	6 4	Footwear; gaiters and the like; parts of such articles	Textiles & Apparel	4 1 . 3 9	1 3 . 9 1	2 7 . 4 8 3
9	1 0	9 5	Toys, games and sports		8 5 . 1 1	5 0 . 8 6	3 4 . 2 4 5

Rank		HS 2		Description	Segment	World(B)		China(B)		Gap(B)										
				requisites; parts and accessories thereof	Toys, Games & Sports															
1	0	1	1	4	2	Articles of leather; saddlery and harness; travel goods, handbags and similar containers; articles of animal gut (other than silk-worm gut)	Bags & Travel Goods	3	2	0	2	1	2	4	7	1	9	5	5	3
1	1	1	2	9	6	Miscellaneous manufactured articles	Seasonal & Festive Goods	2	7	3	2	9	8	1	1	7	5	2	3	
1	2	1	3	5	7	Carpets and other textile floor coverings	Textiles & Apparel	1	1	3	0	1	3	2	9	9	8	1		
1	3	1	4	6	3	Textiles, made up articles; sets; worn clothing and worn textile articles; rags	Textiles & Apparel	2	9	4	0	1	3	5	0	1	5	9	0	4
1	4	1	5	6	9	Ceramic products	Kitchenware & Tableware	1	9	9	9	6	9	7	1	3	0	2	3	
1	5	1	6	6	5	Headgear and parts thereof	Textiles & Apparel	7	2	1	3	9	4	3	2	6	5			
1	6	1	7	4	6	Manufactures of straw, esparto or other plaiting materials; basketware and wickerwork	Seasonal & Festive Goods	0	7	7	0	3	6	0	4	1	4			
1	7	1	8	6	7	Feathers and down, prepared; and articles made of feather or of down; artificial flowers; articles of human hair	Seasonal & Festive Goods	2	1	1	1	6	3	0	4	8	7			
1	8	1	9	6	6	Umbrellas, sun umbrellas,	Textiles & Apparel	1	4	7	1	2	8	0	1	9	8			

Rank	HS 2	Description	Segment	World(B)	China(B)	Gap(B)
		walking-sticks, seat sticks, whips, riding crops; and parts thereof				

Section 6 — Strategic Conclusions & Recommendations

Key Findings

Based on the Canada Import Gap analysis filtered to the **Yiwu small commodity universe**, the strategic conclusions apply:

1 . High-Priority Segments (large gap, low Chinese penetration) These represent the immediate export opportunity — Canada is buying heavily from other countries, and China's share remains the segment average:

- Categories landing in the **PRIORITY quadrant** (upper-left) should be the first focus for Yiwu strategies
- Look for HS 6 product codes with CN share < 20 % AND world market > USD 1 cumulative

2 . Growth Segments (large gap, China already present) China has demonstrated competitiveness but hasn't reached saturation. Opportunity exists to take additional share:

- These require competitive differentiation — price, certification (CSA/UL), or distribution partnership
- Useful for suppliers already exporting to Canada seeking volume growth

3 . Yiwu-Specific Advantages The Yiwu market has structural advantages in:

- **Toys & Games (HS 95)**: deep manufacturing ecosystem, all product types
- **Bags & Leather Goods (HS 42)**: cost leadership, wide SKU variety
- **Home Décor & Hardware (HS 83, 94)**: fast sampling, flexible MOQ
- **Seasonal/Festive (HS 96, 67, 46)**: speed-to-market for trend products

4 . Barriers to Watch

- Canadian product safety standards (CCPSA, Health Canada) for toys and children's items
- CBSA tariff classifications and anti-dumping measures on certain steel/textile products
- Growing preference for certified, sustainably-sourced products among Canadian retailers

Recommended Next Steps

Priority	Action	Target Se
 High	Identify Canadian retail buyers for top-ranked categories	Toys, Hon
 High	Request HS 6 product samples from Yiwu market matching gap products	All segme
 Medium	Evaluate cost of CSA/UL certification for priority HS 6 lines	Toys, Elec
 Medium	Screen for Canadian distribution partners (wholesalers, Amazon FBA)	Bags, Kite
 Low	Monitor tariff changes under CUSMA and Canada–China trade relations	All

Report generated by Mondoré Consulting · Data: UN Comtrade API · Analysis: Canada Import G
Methodology

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```
In [17]: # — Export instructions —
print('='*65)
print('  EXPORT TO PDF')
print('='*65)
print()
print('Option 1 – Command line (requires nbconvert + LaTeX):')
print('  jupyter nbconvert --to pdf yiwu_canada_opportunity.ipynb')
print()
print('Option 2 – HTML then print to PDF (simpler, recommended):')
print('  jupyter nbconvert --to html yiwu_canada_opportunity.ipynb')
print('  Then open the HTML in your browser and Print → Save as PDF')
print()
print('Option 3 – In Jupyter Lab:')
print('  File → Export Notebook As → PDF')
print()
print('Files generated in this run:')
for f in ['fig_01_kpis.png', 'fig_02_top25_gaps.png', 'fig_03_quadrant_all.',
          'fig_04_yiwu_segments.png', 'fig_05_yiwu_quadrant.png',
          'fig_06_segment_detail.png', 'fig_07_hs6_deepdive.png',
          'fig_08_yiwu_ranking.png', 'fig_09_trends.png',
          'yiwu_canada_opportunity.xlsx']:
    exists = '☒' if os.path.exists(f) else '☐'
    print(f'  {exists} {f}')
```

EXPORT TO PDF

Option 1 – Command line (requires nbconvert + LaTeX):

```
jupyter nbconvert --to pdf yiwu_canada_opportunity.ipynb
```

Option 2 – HTML then print to PDF (simpler, recommended):

```
jupyter nbconvert --to html yiwu_canada_opportunity.ipynb
```

Then open the HTML in your browser and Print → Save as PDF

Option 3 – In Jupyter Lab:

File → Export Notebook As → PDF

Files generated in this run:

- ✓ fig_01_kpis.png
- ✓ fig_02_top25_gaps.png
- ✓ fig_03_quadrant_all.png
- ✓ fig_04_yiwu_segments.png
- ✓ fig_05_yiwu_quadrant.png
- ✓ fig_06_segment_detail.png
- ✓ fig_07_hs6_deepdive.png
- ✓ fig_08_yiwu_ranking.png
- ✓ fig_09_trends.png
- ✓ yiwu_canada_opportunity.xlsx